

Sprint Implementation Plan & Engineering Task Breakdown

RetailSync: Centralized Super Shop Warehouse Management System

Subtitle: 14-Week Agile Scrum Roadmap, Sprint Backlogs, Story Points, and Definition of Done

Course: SE-231 (Software System Analysis & Design / Capstone Project 2)

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1. Scrum Framework Overview & Velocity Model

RetailSync follows an **Agile Scrum Development Lifecycle** structured across **six 2-week sprints** and a final **2-week hardening and defense preparation phase** (14 weeks total):

- **Sprint Cadence:** 2 weeks per sprint.
- **Sprint Planning:** Day 1 of each sprint (2 hours).
- **Daily Standup:** Asynchronous or 10-minute sync.
- **Sprint Review & Demo:** Alternate Friday with academic supervisor / evaluators.
- **Definition of Done (DoD):** Code committed to Git, unit tests passing (> 80% coverage), API endpoints documented in OpenAPI, database migrations reproducible via Docker, and UI responsive on mobile viewport (375x667).

14-WEEK CAPSTONE IMPLEMENTATION ROADMAP	
Sprint 1 (W01-W02) Database Normalization, Core DDL Migrations, Auth & Next.js PWA Scaffolding	
Sprint 2 (W03-W04) Inbound PO Lifecycle, Handheld Barcode Dock Scanning, and Digital GRN Generation	
Sprint 3 (W05-W06) Directed Spatial Putaway Engine, 2D Floor Map & Real-Time FEFO Inventory Ledger	
Sprint 4 (W07-W08) Sub-2.0s POS Concurrency Deduction Engine, Row-Locking & Offline Caching Queue	
Sprint 5 (W09-W10) Replenishment DSS (EOQ, Greasley Safety Stock, ROP) & Outbound Wave Store Dispatch	
Sprint 6 (W11-W12) Blind Cycle Counting, Shrinkage Anomaly Detection & Executive Floor Telemetry	
Sprint 7 (W13-W14) Concurrency Stress-Testing (Locust 50 req/s), Security Audit, Capstone Viva Demo	

2. Sprint Backlogs & Granular Task Breakdowns

2.1 Sprint 1 (Weeks 1-2): Architecture, Relational Schema & Auth Core

- **Sprint Goal:** Establish the reproducible Docker environment, initialize the 3NF PostgreSQL database, and deliver secure JWT authentication with role-based access control.
- **Target Velocity:** 32 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-101	Setup Docker Compose orchestrating PostgreSQL 16, Redis 7, FastAPI backend, and Next.js frontend.	DevOps & Infra	5	Single command <code>docker compose up -d</code> boots all containers with healthy status.
TSK-102	Implement Alembic migration scripts for 15 core database tables and composite B-Tree indexes.	Data Modeling	8	<code>alembic upgrade head</code> runs cleanly; database schema passes 3NF verification audit.
TSK-103	Load initial authentic seed data (5 suppliers, 7 SKUs, 6 bins, 6 batches, 3 stores).	Data Modeling	3	Seed script populates DB; verify foreign key constraints prevent orphaned records.
TSK-104	Develop User entity, Argon2id password hashing, and JWT token issuance (<code>/api/v1/auth/login</code>).	Security & Auth	5	Pytest confirms valid token issued; invalid credentials return HTTP 401.
TSK-105	Implement RBAC middleware enforcing permissions for Cashier, Operator, Clerk, Supervisor, Procurement.	Security & Auth	5	Test unauthorized route access returns <code>HTTP 403 Forbidden</code> .
TSK-106	Scaffold Next.js 14 frontend design system with Tailwind CSS and responsive high-contrast layout.	Frontend PWA	6	Dashboard renders cleanly on mobile screen (375px) and desktop monitor (1920px).

2.2 Sprint 2 (Weeks 3–4): Inbound PO & Handheld Barcode Dock Receiving

- **Sprint Goal:** Deliver working dock receiving workflow allowing clerks to scan delivered carton barcodes against open POs and commit digital GRNs with damaged quarantine handling.
- **Target Velocity:** 34 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-201	Develop Supplier and Purchase Order CRUD API endpoints (<code>/api/v1/suppliers</code> , <code>/api/v1/inbound/po</code>).	Inbound Receiving	5	OpenAPI docs render PO status lifecycle transitions cleanly.

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-202	Integrate Html5-QRCode / ZXing camera and Bluetooth scanner input in Next.js PWA.	Mobile Scanning	8	Scanner decodes EAN-13 barcodes in < 350 ms with audio beep feedback.
TSK-203	Build PO line-item scan verification endpoint with remaining shelf-life check ($\geq 75\%$).	Inbound Receiving	5	Batches with < 75% shelf life are flagged with visual rejection banner.
TSK-204	Implement damaged carton logging with optional photo upload and quarantine status assignment.	Inbound Receiving	5	Quarantined units saved to product_batches with quarantine_status='QUARANTINED' .
TSK-205	Build atomic GRN completion transaction generating digital GRN and supplier credit note advisory.	Inbound Receiving	8	Commits GRN; creates inventory_transactions records; updates PO to COMPLETED .
TSK-206	Design printable internal warehouse carton/pallet barcode labels (EAN-128 format).	Inbound Receiving	3	Label layout prints cleanly on 4x6 inch thermal receipt/label printer.

2.3 Sprint 3 (Weeks 5–6): Directed Putaway & Real-Time FEFO Inventory Ledger

- **Sprint Goal:** Enable guided pallet putaway into warehouse bins and enforce automated First-Expired, First-Out (FEFO) batch priority allocation.
- **Target Velocity:** 35 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-301	Implement warehouse spatial bin layout API (locations) with zone and capacity constraints.	Spatial Putaway	5	Database rejects putaway suggestions into bins exceeding 500 kg or 300 units.
TSK-302	Develop Directed Putaway recommendation algorithm factoring SKU velocity (ABC) and temperature.	Spatial Putaway	8	Class-A chilled milk correctly routed to Chilled Zone Aisle 1 near dispatch dock.

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-303	Build mobile Directed Putaway PWA screen requiring destination bin scan confirmation.	Mobile Scanning	5	Operator cannot commit putaway without scanning the correct target bin barcode.
TSK-304	Develop Real-Time FEFO Priority Queue sorting batches by ascending expiry_date .	FEFO Engine	8	API test verifies stock allocation always selects nearest valid expiry batch.
TSK-305	Implement automated Expiry Watchlist background job (Safe, Watch, Warning, Critical ≤ 14 days).	FEFO Engine	5	Batches reaching expiration are automatically locked to quarantine_status='EXPIRED' .
TSK-306	Build interactive 2D spatial bin occupancy floor map in Next.js using CSS Grid / Canvas.	Frontend Visual	4	Bins color-code dynamically: Green (< 70%), Amber (70-90%), Red (> 90%).

2.4 Sprint 4 (Weeks 7-8): Sub-2.0s POS Sync API & Concurrency Row-Locking

- **Sprint Goal:** Build and stress-test the real-time POS stock deduction API, guaranteeing sub-2.0s response times and zero deadlocks across multiple registers, backed by offline client caching.
- **Target Velocity:** 36 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-401	Implement /api/v1/pos/sync with pessimistic row-level locking (SELECT ... FOR UPDATE).	POS Concurrency	8	Automated benchmark demonstrates database lock duration < 25 milliseconds.
TSK-402	Implement X-Idempotency-Key token validation in Redis with 48-hour TTL.	POS Concurrency	5	Replaying duplicate POST request returns cached confirmation without double-deduction.
TSK-403	Develop POS register registry and telemetry heartbeat tracking online/offline status.	POS Concurrency	5	Dashboards highlight disconnected store cash registers in real time.

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-404	Implement client-side Service Worker and encrypted IndexedDB offline sales queue.	Offline Resilience	8	Disconnecting network allows 20 scans locally; auto-syncs upon link reconnect.
TSK-405	Develop bulk synchronization endpoint (/api/v1/pos/bulk-sync) for network recovery replay.	Offline Resilience	5	Bulk sync correctly processes 50 buffered offline transactions in order.
TSK-406	Build Locust load-test script simulating 10 POS cashiers competing for the last unit of milk.	QA & Testing	5	100% of race conditions resolve cleanly; zero negative inventory balances occur.

2.5 Sprint 5 (Weeks 9–10): Replenishment DSS & Outbound Wave Dispatch

- **Sprint Goal:** Implement mathematical inventory decision support models (EOQ, Greasley Safety Stock, ROP) and multi-branch outbound store wave picking.
- **Target Velocity:** 33 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-501	Implement Dynamic EOQ formula module using historical annual demand and holding costs.	Decision Support	5	Computed EOQ matches manual mathematical formula verification within 0.01%.
TSK-502	Implement Greasley's Statistical Safety Stock formula factoring daily demand and vendor lead-time variance.	Decision Support	8	Output accounts for configured Z-scores (90%, 95%, 99%) and vendor \sigma_L.
TSK-503	Build dynamic ROP alert engine triggering automated draft Purchase Orders for procurement review.	Decision Support	5	Falling below ROP generates pre-filled draft PO ready for one-click approval.
TSK-504	Develop Branch Store Requisition Portal allowing store managers to submit weekly orders.	Store Requisition	5	Store order displays real-time central warehouse stock availability.

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-505	Build Wave Picking consolidation algorithm aggregating multiple store requisitions into one wave.	Outbound Picking	5	Aggregates duplicate SKU requests into single bulk pick task.
TSK-506	Implement shortest-path pick-list routing ordering picker travel by Aisle and Shelf sequence.	Outbound Picking	5	Picker walking distance reduced by > 40% compared to unsequenced list.

2.6 Sprint 6 (Weeks 11-12): Cycle Counting, Anomaly Detection & BI Dashboards

- **Sprint Goal:** Implement continuous blind cycle counting, unsupervised ML shrinkage anomaly detection, and executive KPI analytics.
- **Target Velocity:** 30 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-601	Implement Blind Cycle Counting sheets (system quantities hidden from floor counters).	Stocktaking & Audit	5	Mobile UI requires manual number input; reveals discrepancy only after submission.
TSK-602	Build supervisor discrepancy approval workflow with mandatory reason codes and photo proofs.	Stocktaking & Audit	5	Adjustments > 1,000 BDT require supervisor authorization before ledger commit.
TSK-603	Implement Scikit-learn IsolationForest model to detect anomalous shrinkage clusters.	Machine Learning	8	Model flags synthetic theft clusters across specific shift operators (F1 > 0.85).
TSK-604	Build Executive KPI Dashboard (Total Inventory Value, GMROI, Turnover Ratio, Spoilage Rate).	Executive BI	6	Dashboard renders historical trends using Chart.js / Recharts with sub-1.5s load.
TSK-605	Enforce database-level immutable audit ledger trigger on inventory_transactions .	Security & Audit	6	Direct SQL DELETE or UPDATE statements are blocked by database trigger exception.

2.7 Sprint 7 (Weeks 13–14): Concurrency Stress-Testing, UAT & Viva Defense

- **Sprint Goal:** Conduct comprehensive load testing (50 req/s), complete end-to-end user acceptance testing (UAT), and prepare the interactive defense presentation.
- **Target Velocity:** 25 Story Points

Task ID	Task Description	Epic	Est. Points	DoD Verification Step
TSK-701	Execute high-throughput Locust stress test (50 concurrent scanners, 10 POS registers).	Stress Testing	8	99th percentile response time remains < 1.5 seconds with zero database errors.
TSK-702	Conduct OWASP ZAP security vulnerability scan on all authentication and API endpoints.	Security Audit	5	Zero Critical or High vulnerabilities identified; SSL Labs A+ rating verified.
TSK-703	Execute complete end-to-end UAT scenario covering receiving dock to retail POS checkout.	System Testing	5	All 6 personas execute workflows seamlessly on test hardware.
TSK-704	Finalize Capstone Project Proposal Part 1 & Part 2 documentation and interactive slides.	Academic Defense	7	Deliverables compiled to PDF; interactive showcase dashboard deployed online.